GROUP 2

# FINAL REPORT

A presentation about what our team has been doing on data science projects over the past period

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# ABOUT US

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**Nguyễn Đặng Hoàng Dinh**

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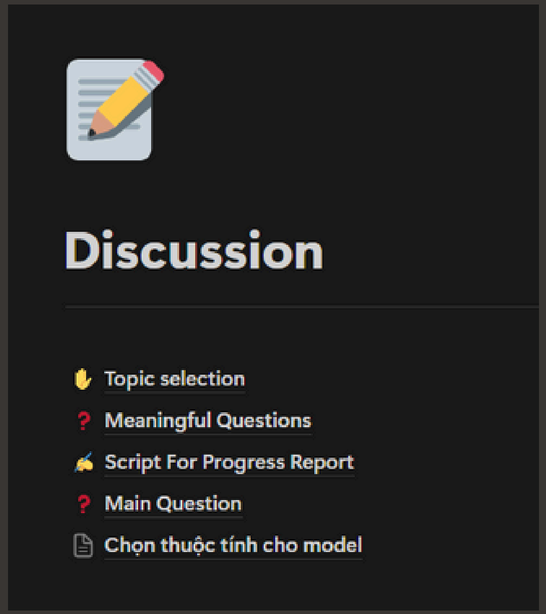
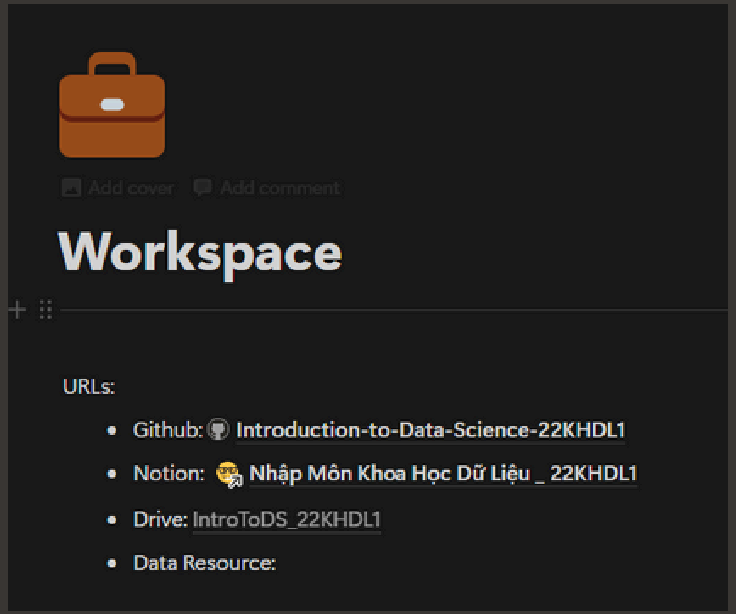
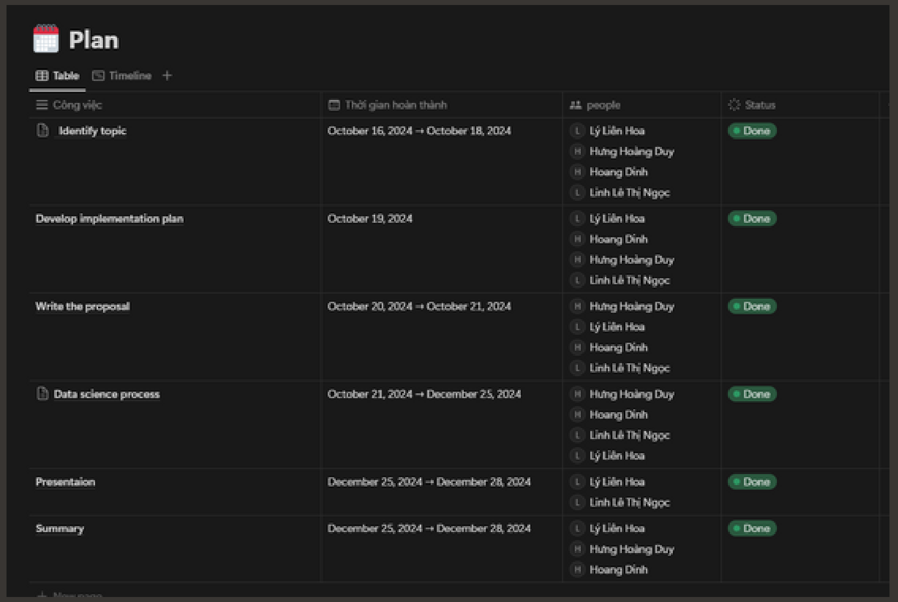
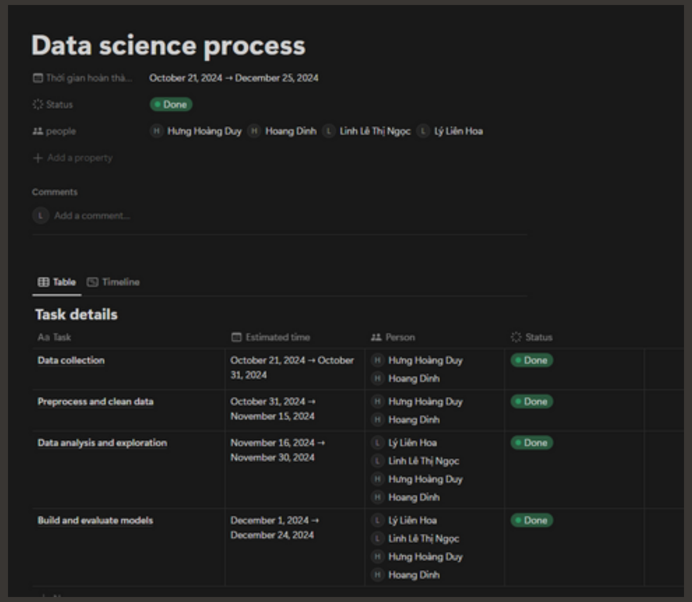
# HOW WE COLLABORATE ?

## Plan & Schedule

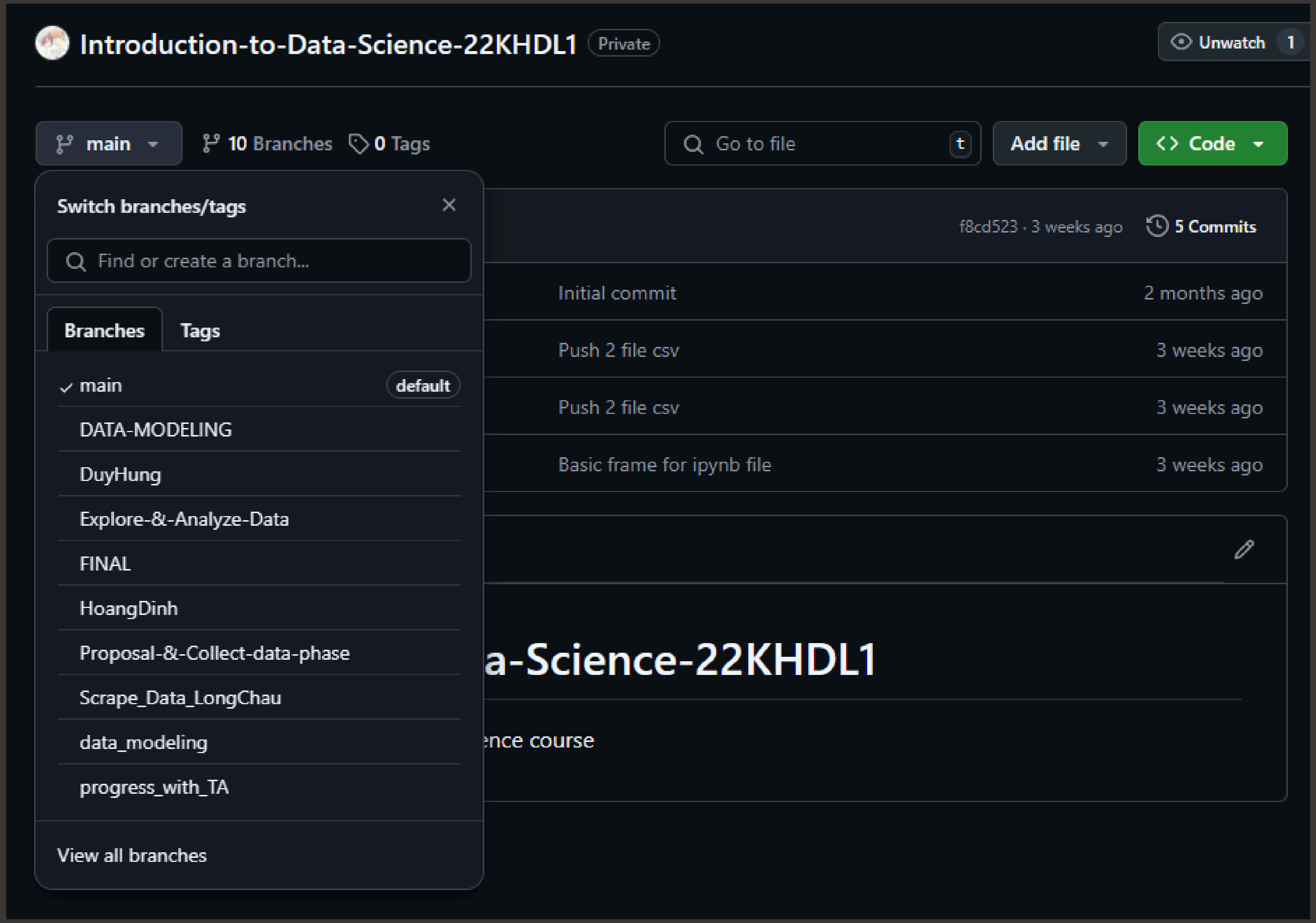



# HOW WE COLLABORATE ?

## Version Control & Interact



# DEFINING THE PROBLEM

NHÀ THUỐC
FPT Retail LONG CHÂU

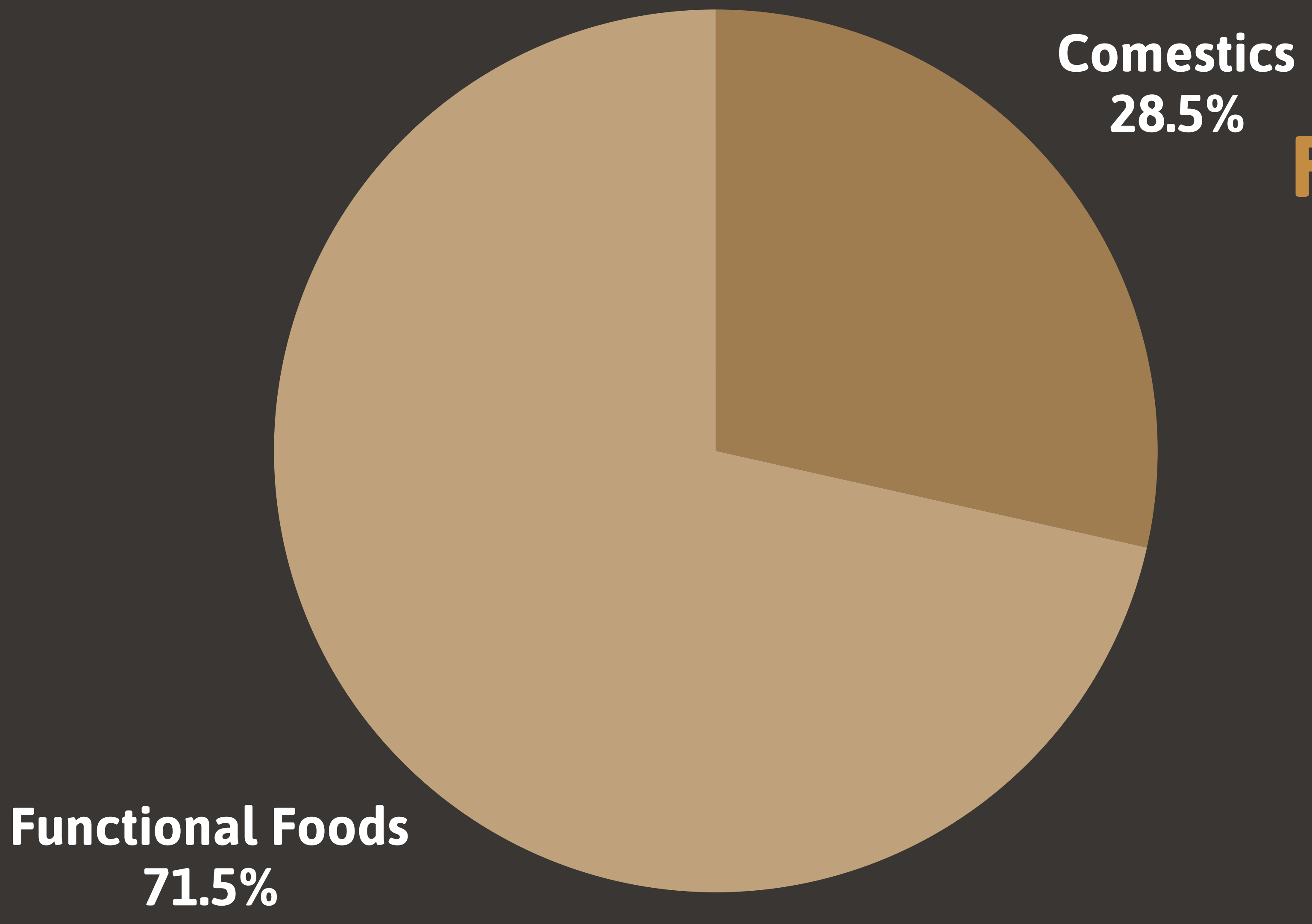
Cusomer needs

**Based on Long Chau's supply, what factors affect cusomers demand?**

# DATA COLLECTION & PREPARATION

**Raw Data**

1999 samples




**Functional foods Data**

1429 samples

**Comestics Data**

570 samples

# DATA PREPROCESSING

## Meaning Of Rows & Columns

| 1 | Product Name | Category | Dosage Form | Price | Trademark | Brand Origin | Country | Rating |
|---|---|---|---|---|---|---|---|---|
| 2 | Gel chấm mụn, mờ thâm Decumar Advance THC 20g dưỡng ẩm da, da mềm, mịn | Chăm sóc cơ thể | Gel | 105000.0 | DECUMAR | Việt Nam | Việt Nam | 5.0 |
| 3 | Dung dịch vệ sinh vùng kín Bimunica 250ml dành cho bé gái từ 0 tháng tuổi | Chăm sóc cơ thể | | 230000.0 | | Hoa Kỳ | Liên Bang Nga | 5.0 |
| 4 | Kem giảm thâm vùng nách, mông, bikini Neothera Armpit Cream La Beaute 35ml dưỡng trắng da | Chăm sóc cơ thể | | 139000.0 | La Beauty | Việt Nam | Việt Nam | 5.0 |
| 5 | Gel rửa mặt SVR Sebiaclear Gel Moussant 200ml làm sạch không chứa xà phòng và loại bỏ tế bào da chết | Chăm sóc cơ thể | | 390000.0 | SVR | Pháp | Pháp | |
| 6 | Bọt vệ sinh nam giới Sumely Men's Sanitary Foam giúp làm sạch, khử và ngăn chặn mùi vùng kín (100ml) | Lăn khử mùi, xịt khử mùi | Dạng bọt | 96000.0 | | Việt Nam | Việt Nam | 5.0 |
| 7 | Sữa rửa mặt On: The Body Rice Therapy Heartleaf Acne Cleanser làm sạch sâu không gây khô da (150ml) | Sữa rửa mặt (Kem, gel, sữa) | | 132000.0 | | Hàn Quốc | Hàn Quốc | 5.0 |
| 8 | Nước tẩy trang JMsolution Water Luminous S.O.S Ringer Cleansing Water giúp làm sạch bụi bẩn, dầu thừa và lớp trang điểm (500ml) | Nước tẩy trang, dầu tẩy trang | | 199000.0 | | Hàn Quốc | Hàn Quốc | 5.0 |
| 9 | Gel Rosette Gommage Clear Peel tẩy tế bào chết (180g) | Tẩy tế bào chết | Gel | 185000.0 | Rosette | Nhật Bản | Nhật Bản | 5.0 |
| 10 | Gel làm mờ sẹo Fixderma Scars do mụn, bỏng, rạn da, vết thươngl (7ml) | Trị sẹo, mờ vết thâm | | 132800.0 | Fixderma | Mỹ | Ấn Độ | 5.0 |
| 11 | Gội phủ bạc Healthy and lively H4 Lavox màu nâu cà phê (10 gói x 30ml) | Dầu gội dầu xả | Dung dịch | 22400.0 | LAVOX | | Việt Nam | 5.0 |

First 10 rows in origin data

# DATA PREPROCESSING

## Handling Missing Data

- **Check percent of NULL values of each column except rating**
  - If the percent **under or equal to 75%** → Skip
  - If the percent **above 75%**
    - Non-numeric → Frequency values
    - Numeric → Median

# DATA PREPROCESSING

## Data Standardization

**“Rating” column**

★★★★★

NULL values

“unknown” values

# DATA PREPROCESSING

## Data Standardization

***"Brand Origin"* Feature**

- Hoa Kỳ
- Mỹ

***"Country"* Feature**

- Hoa Kỳ
- USA

**MỸ**

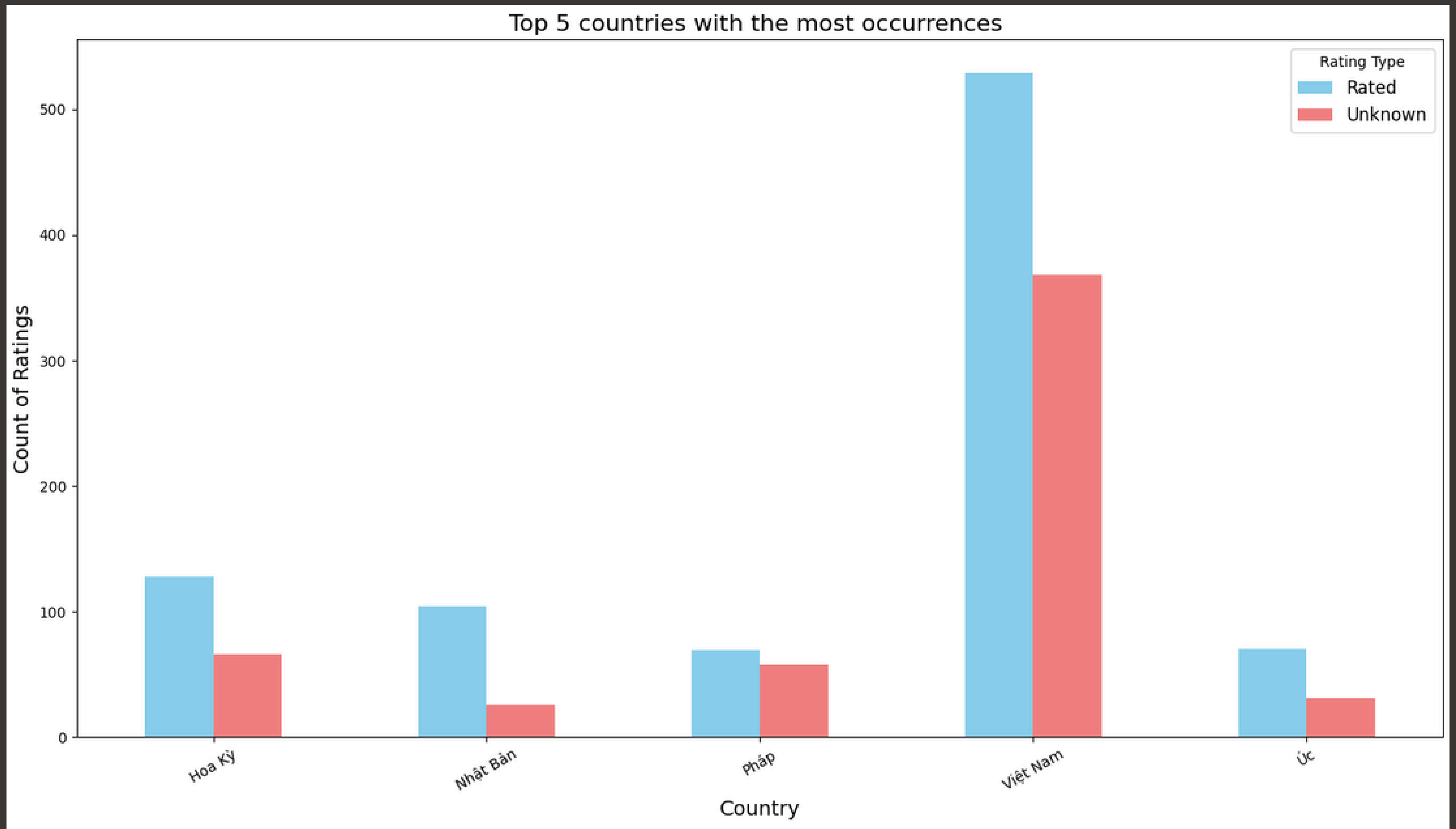
# DATA EXPLORATION & PREPROCESSING

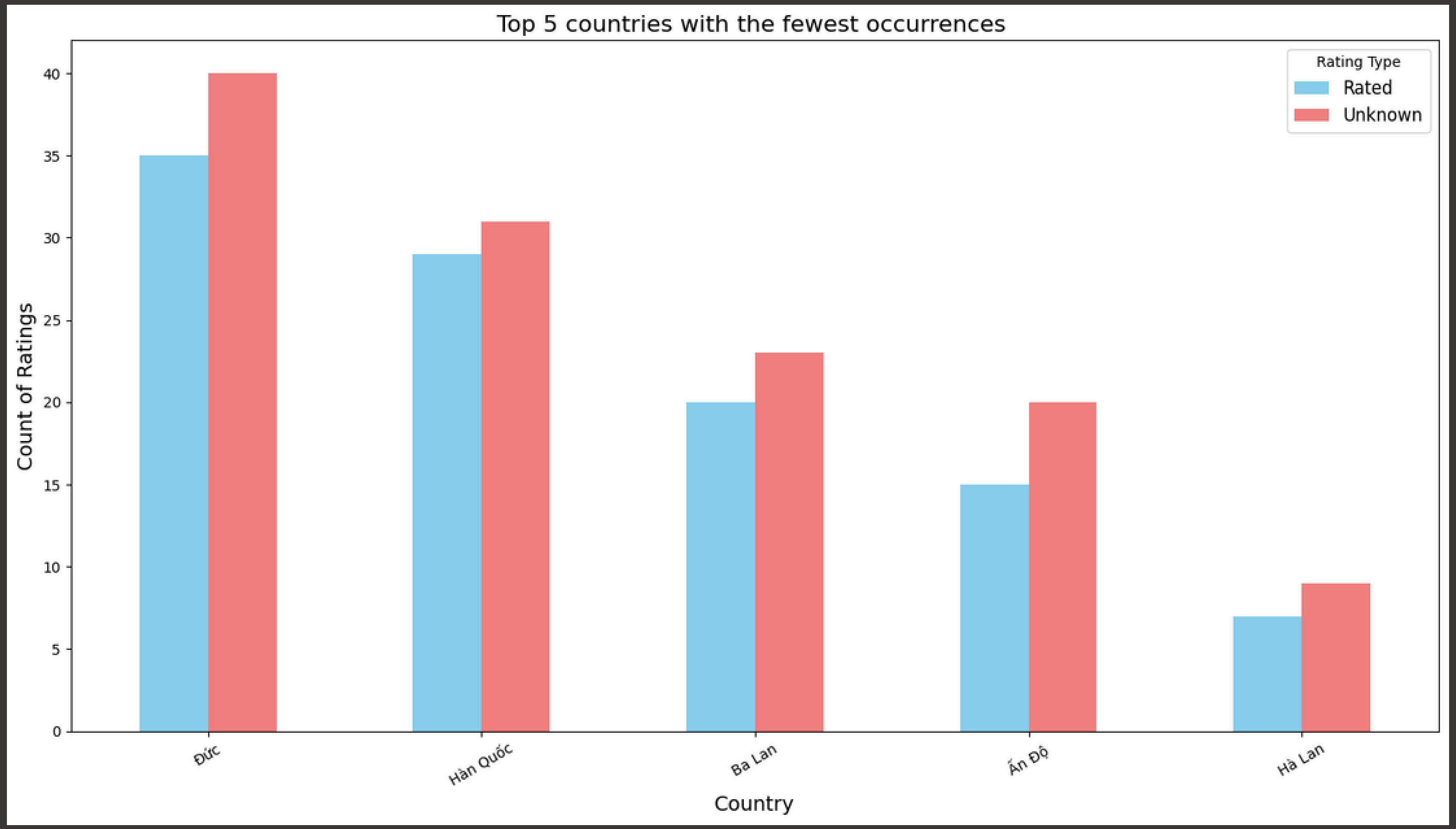
## Questions

1. In each category, which factor is decisive?
2. In each category, which domestic and foreign products receive more attention?
3. In each category, the distribution of prices.
4. In each category, the distribution of brands.
5. In each category, the distribution of dosage forms.

# DATA EXPLORATION & PREPROCESSING

## Question 1: In each category, which factor is decisive?

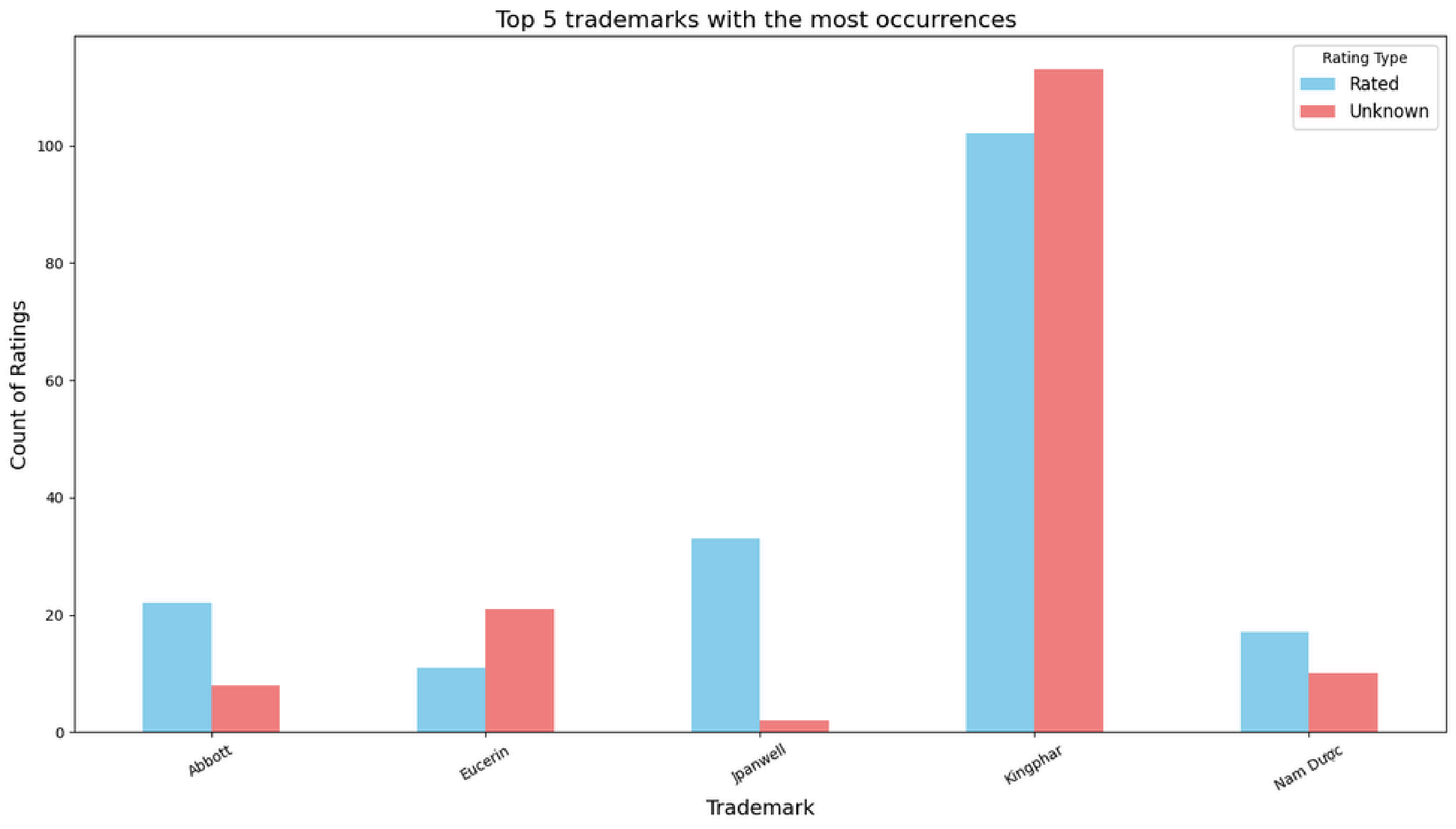


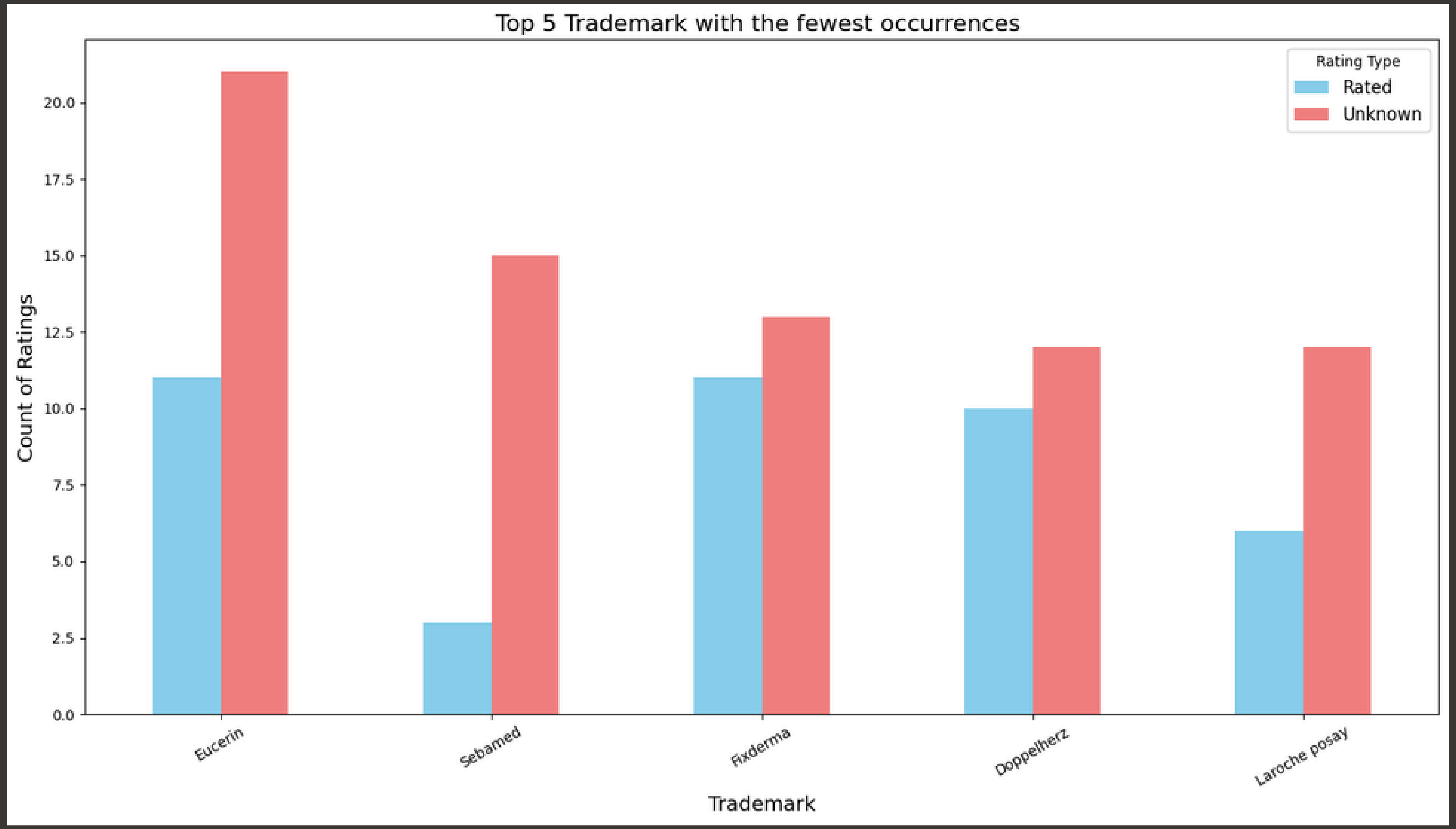




Country

# DATA EXPLORATION & PREPROCESSING

## Question 1: In each category, which factor is decisive?

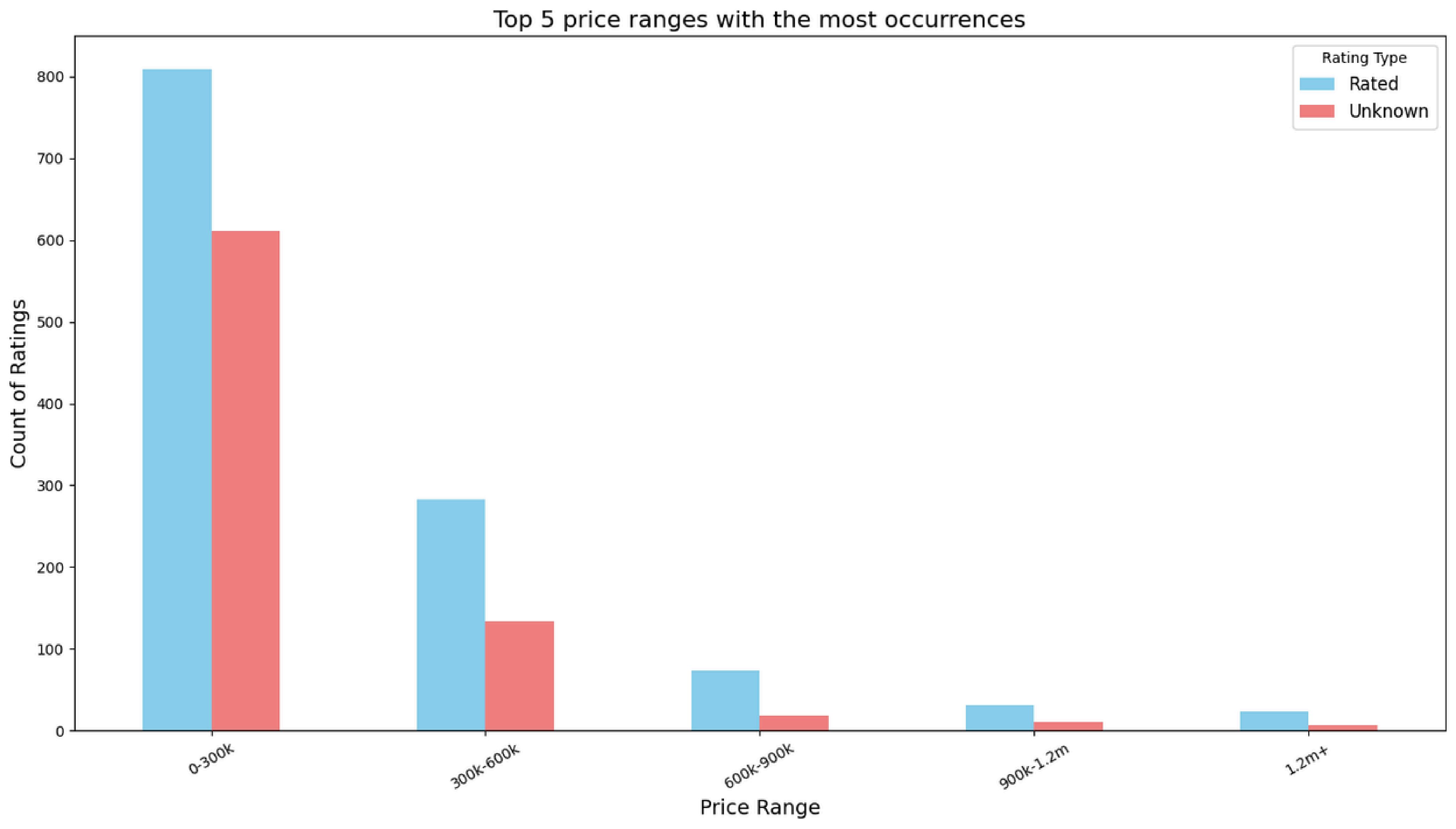



Trademark

# DATA EXPLORATION & PREPROCESSING

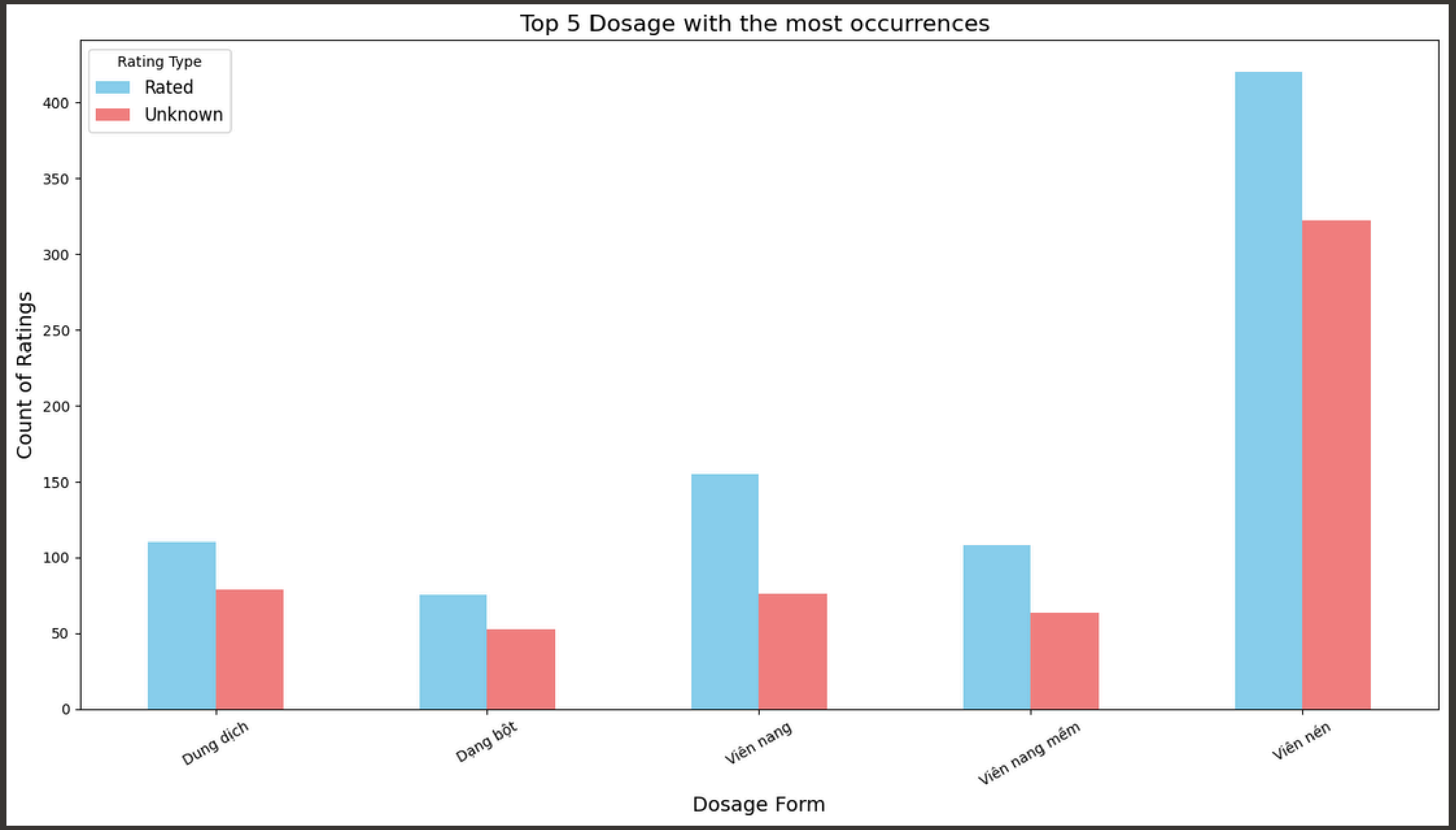
## Question 1: In each category, which factor is decisive?

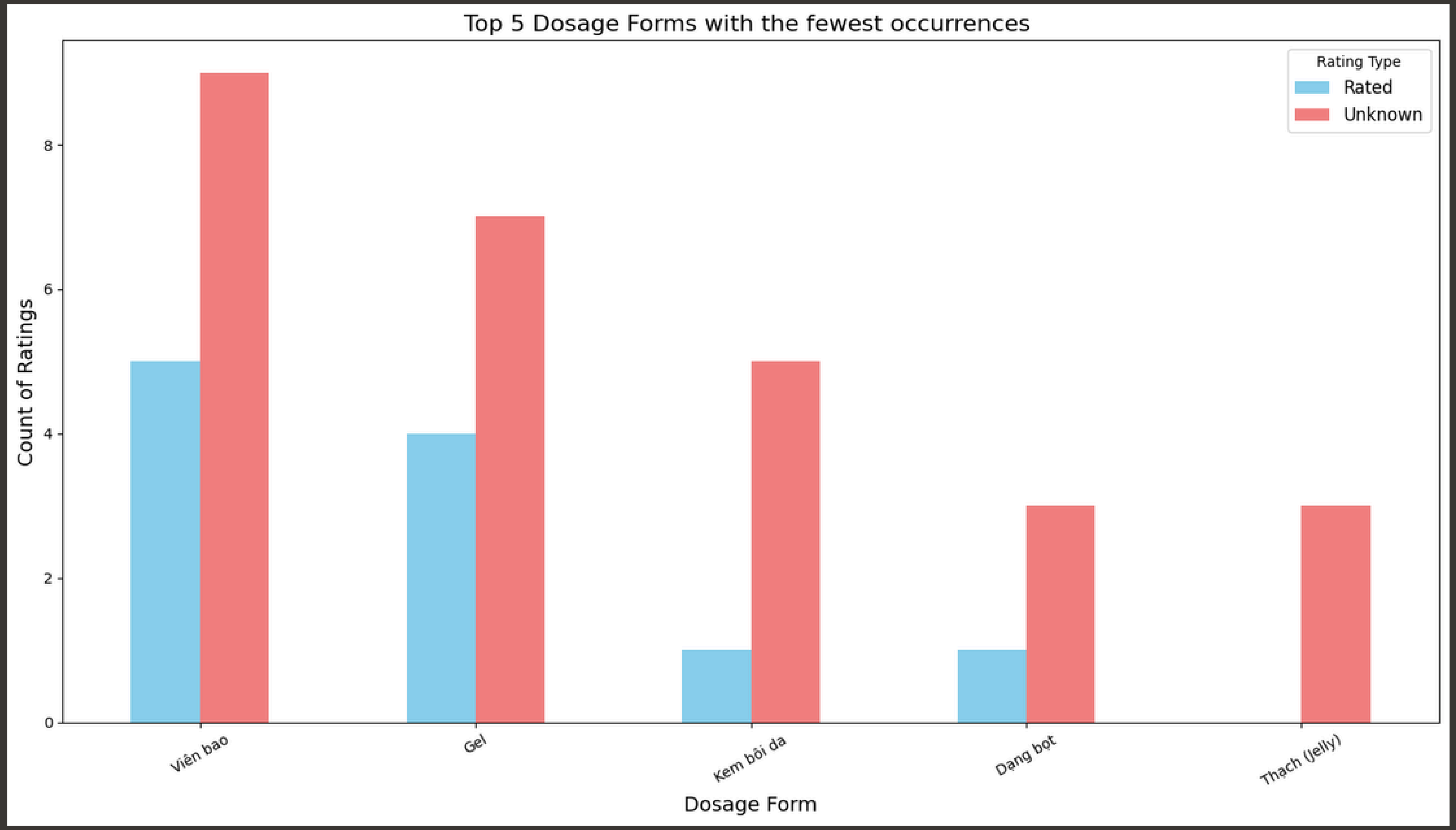



Price

# DATA EXPLORATION & PREPROCESSING

## Question 1: In each category, which factor is decisive?



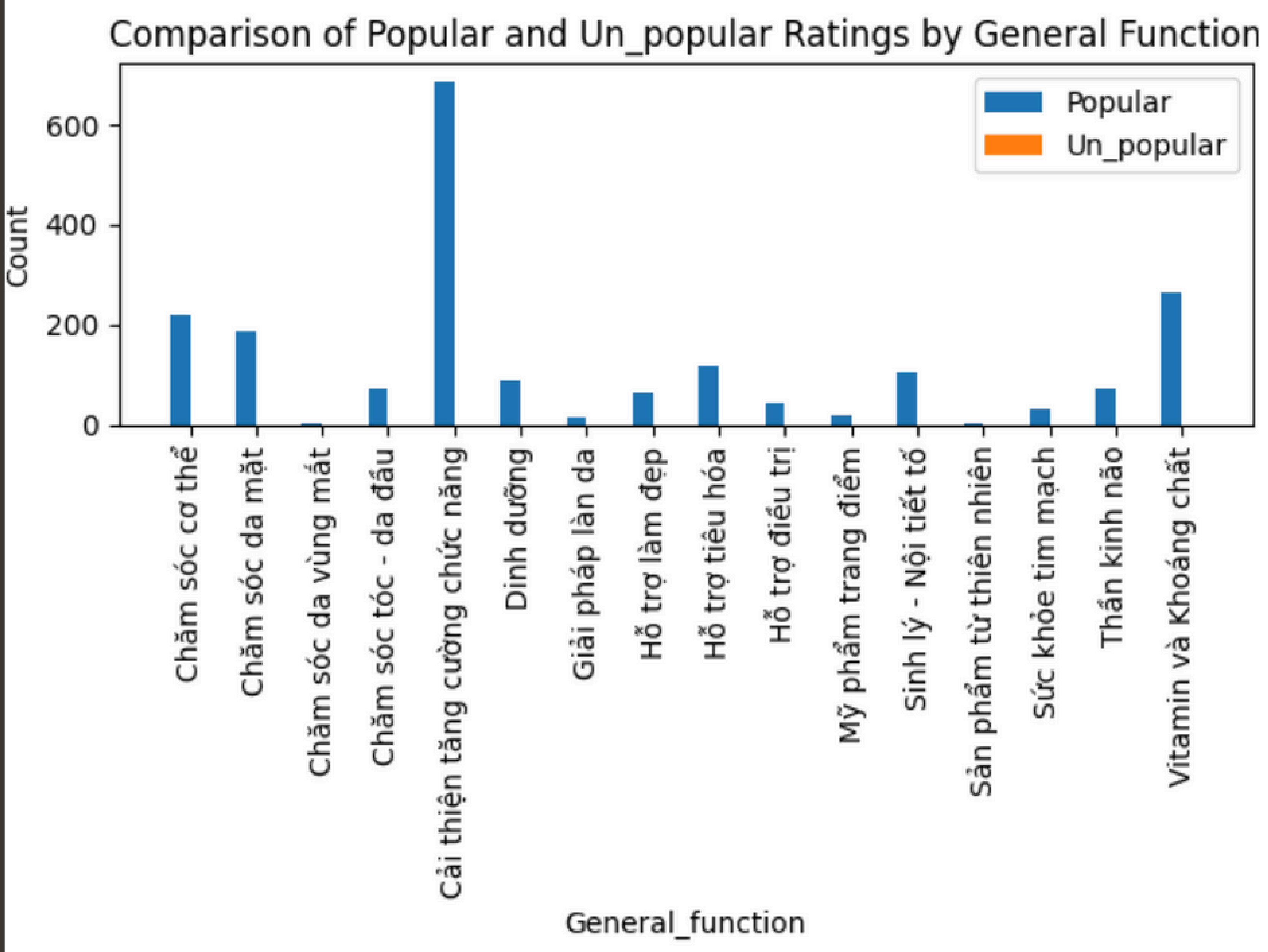



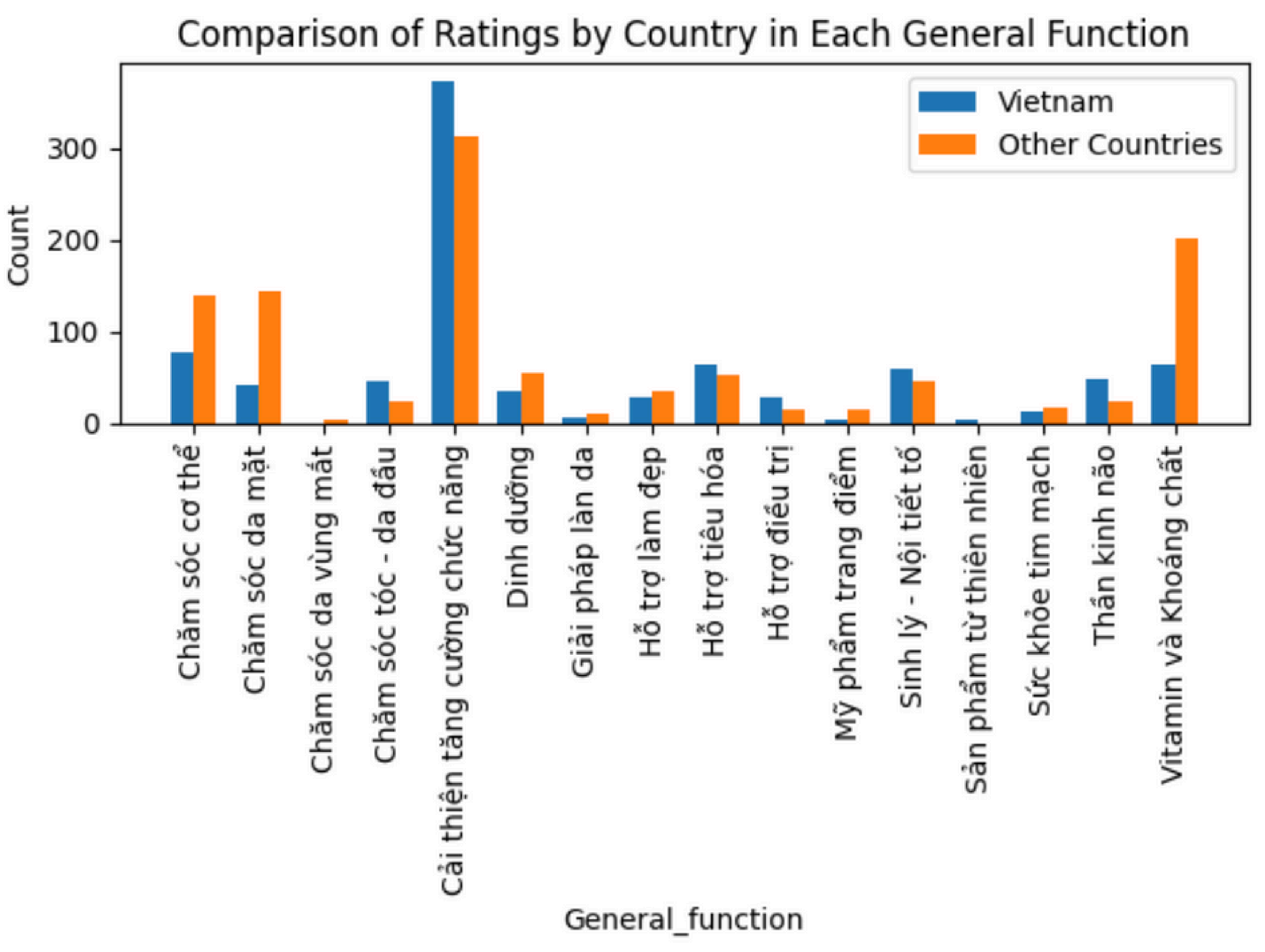

Dosage Form

# DATA EXPLORATION & PREPROCESSING

## Question 2

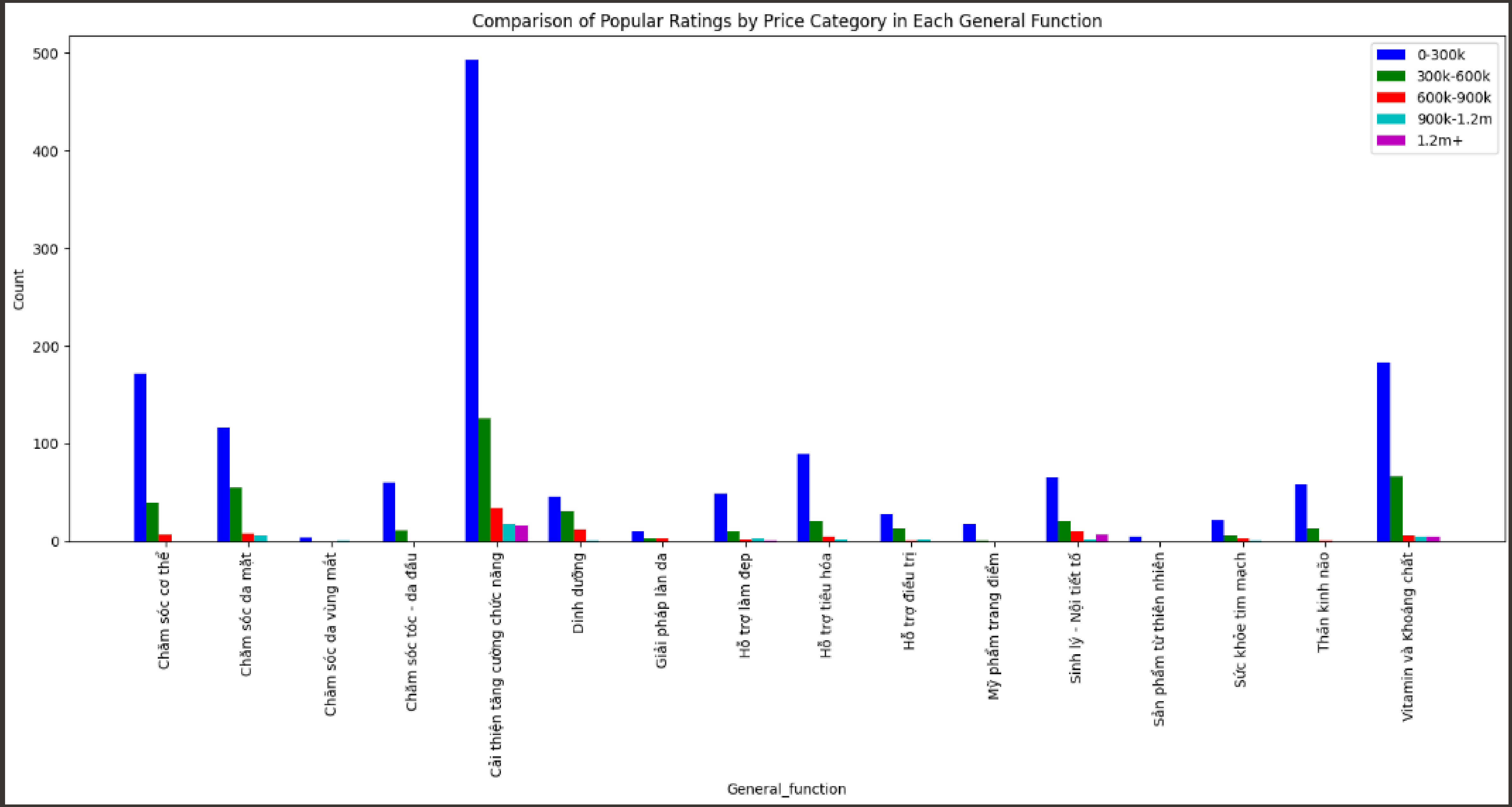
In each category, which domestic and foreign products receive more attention?



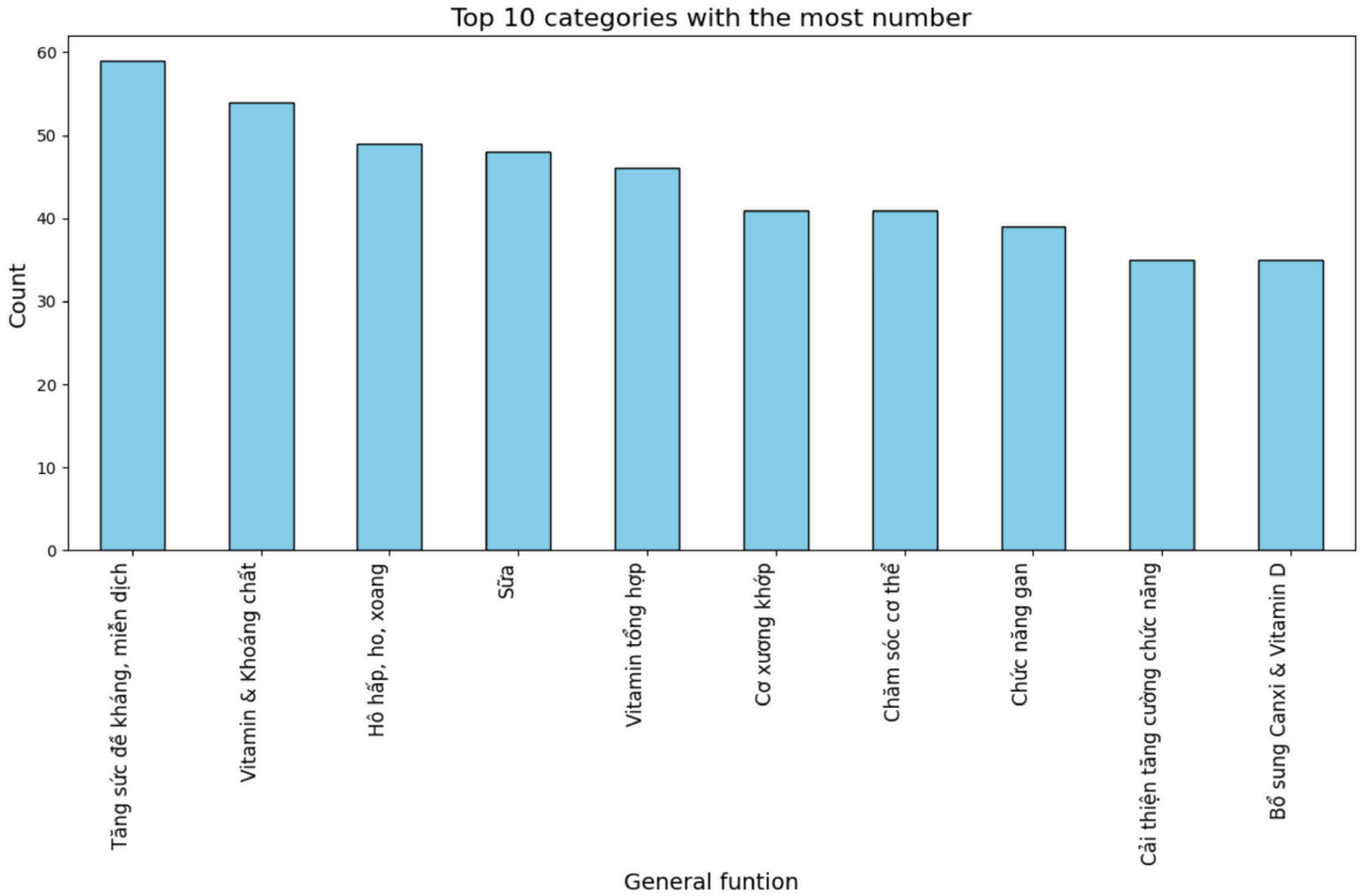
# DATA EXPLORATION & PREPROCESSING

## Question 3 In each category, the distribution of prices.



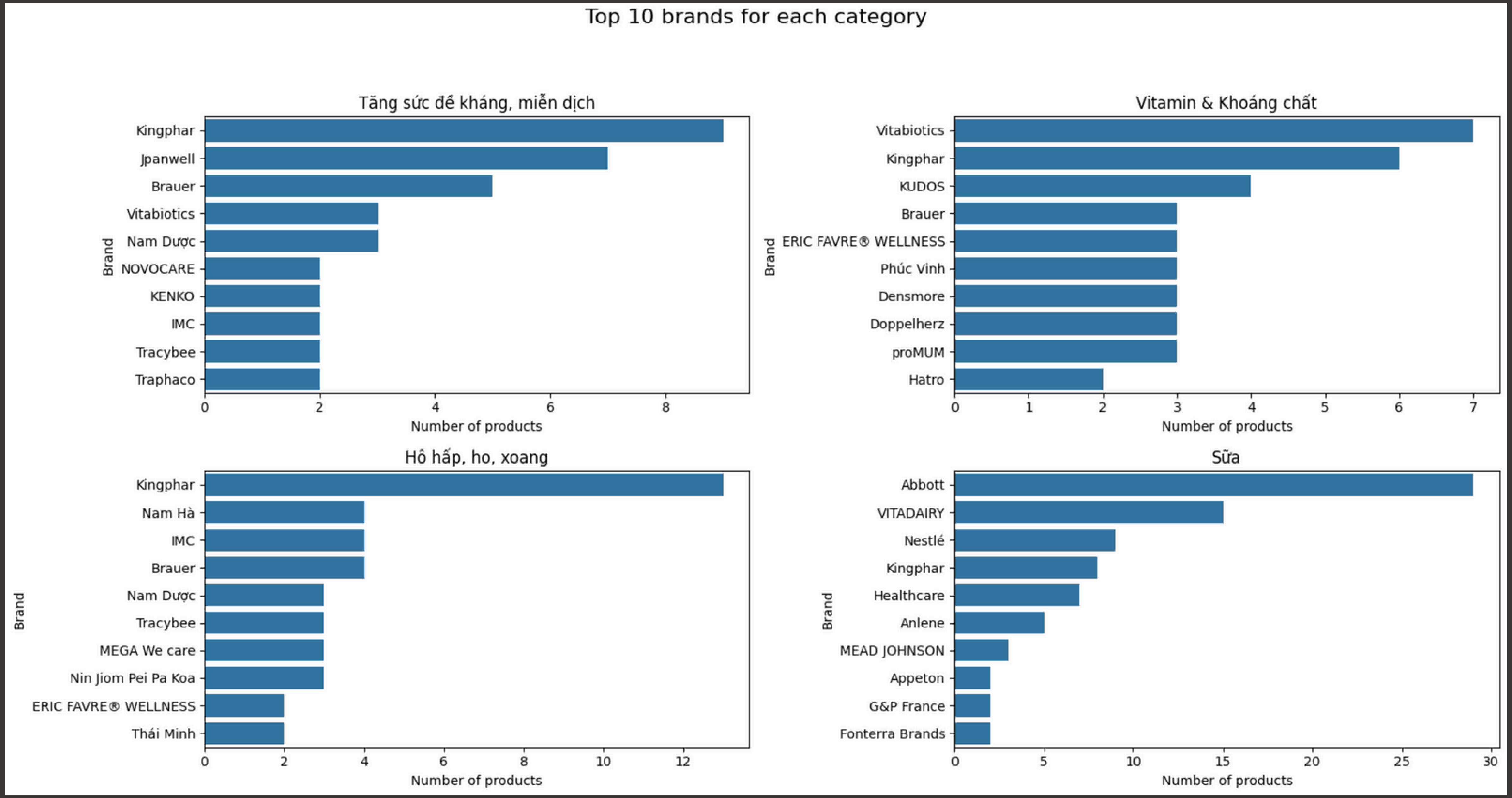
# DATA EXPLORATION & PREPROCESSING

## Question 4 In each category, the distribution of brands.



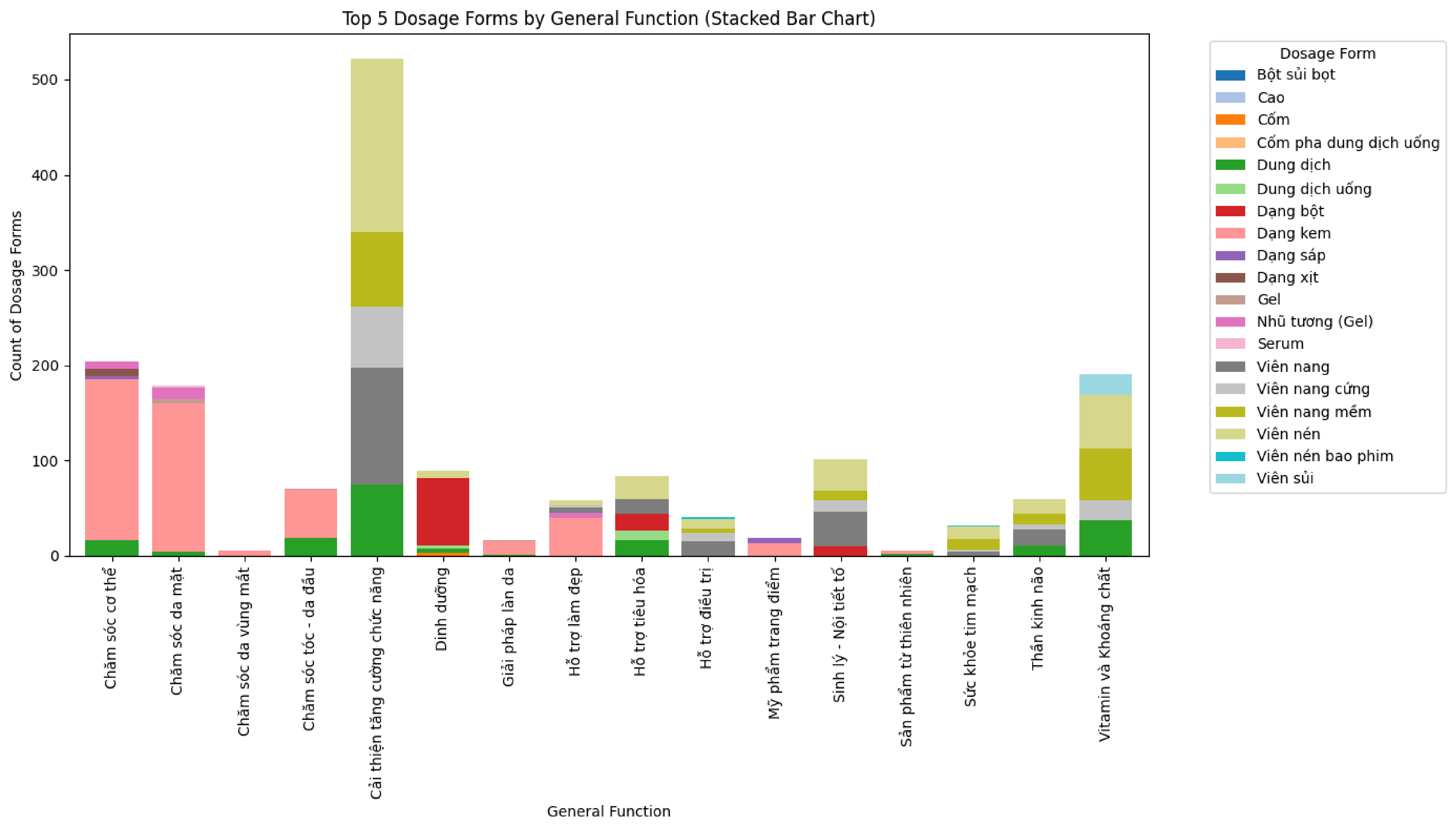
# DATA EXPLORATION & PREPROCESSING

## Question 4 In each category, the distribution of brands.



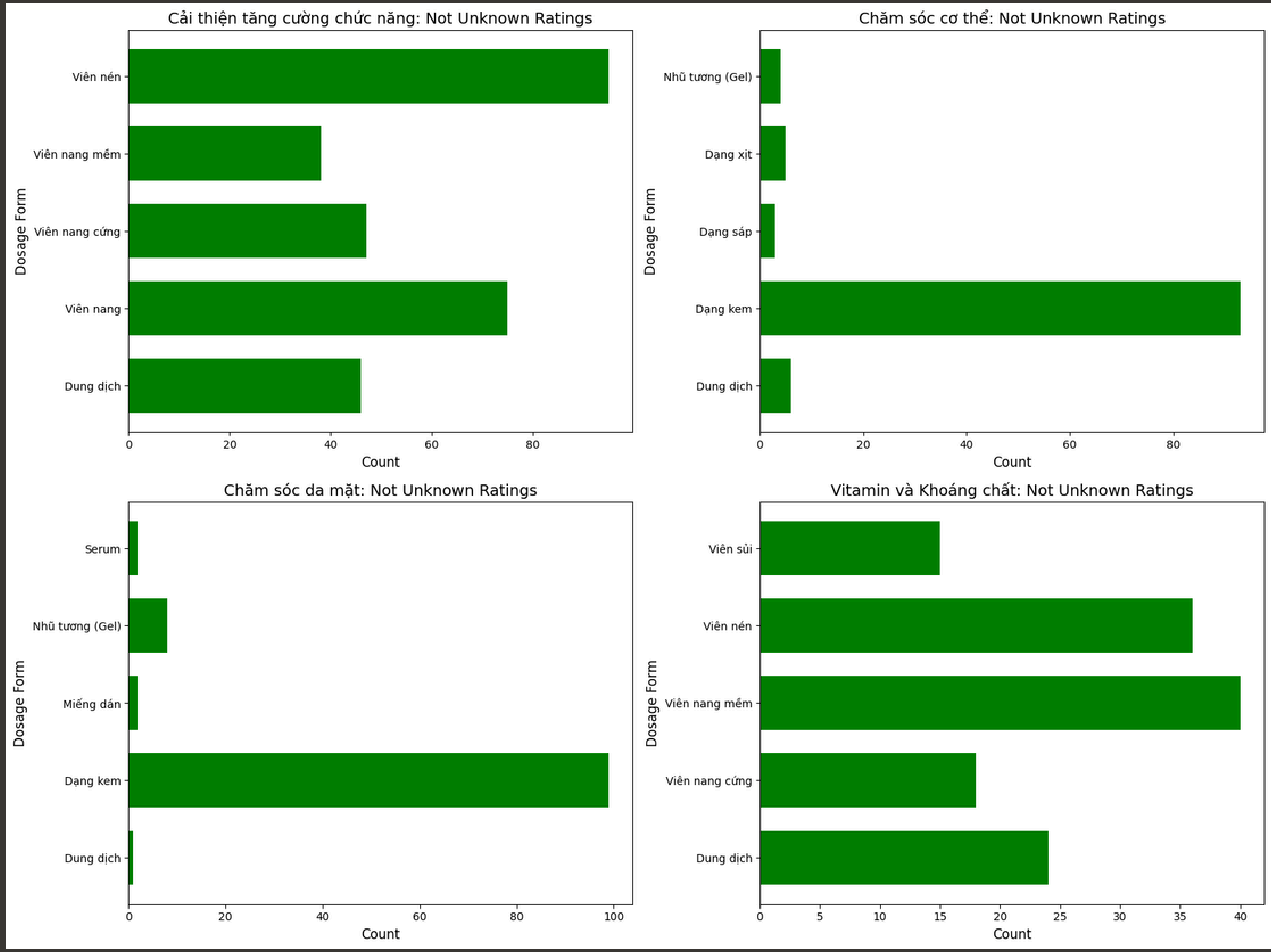
# DATA EXPLORATION & PREPROCESSING

## Question 5 In each category, the distribution of dosage forms.



# DATA EXPLORATION & PREPROCESSING

## Question 5 In each category, the distribution of dosage forms.



# MODEL BUILDING
## Ways To Define Models

**Considered criterias**

Type of problems

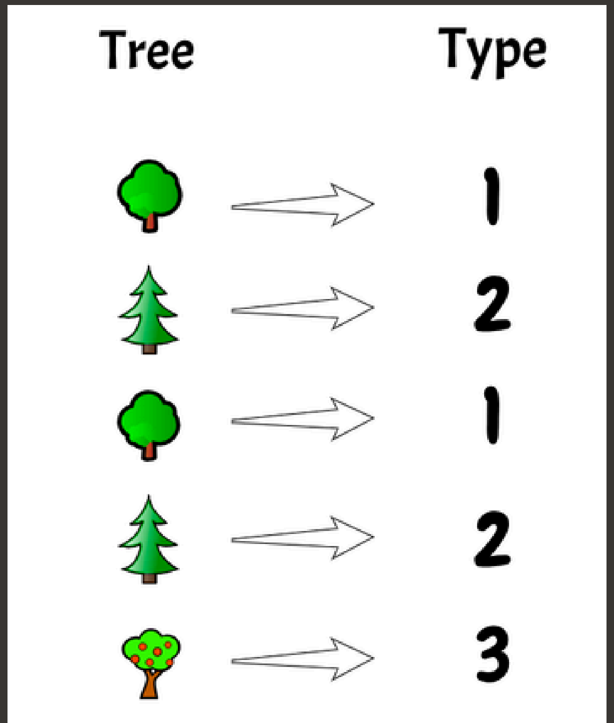
Scale of the data

- Random Forest Regression
- Gradient Boosting (XGBoost)
- Ridge Regression
- SVR(kernel=rbf)
- ElasticNet Regression

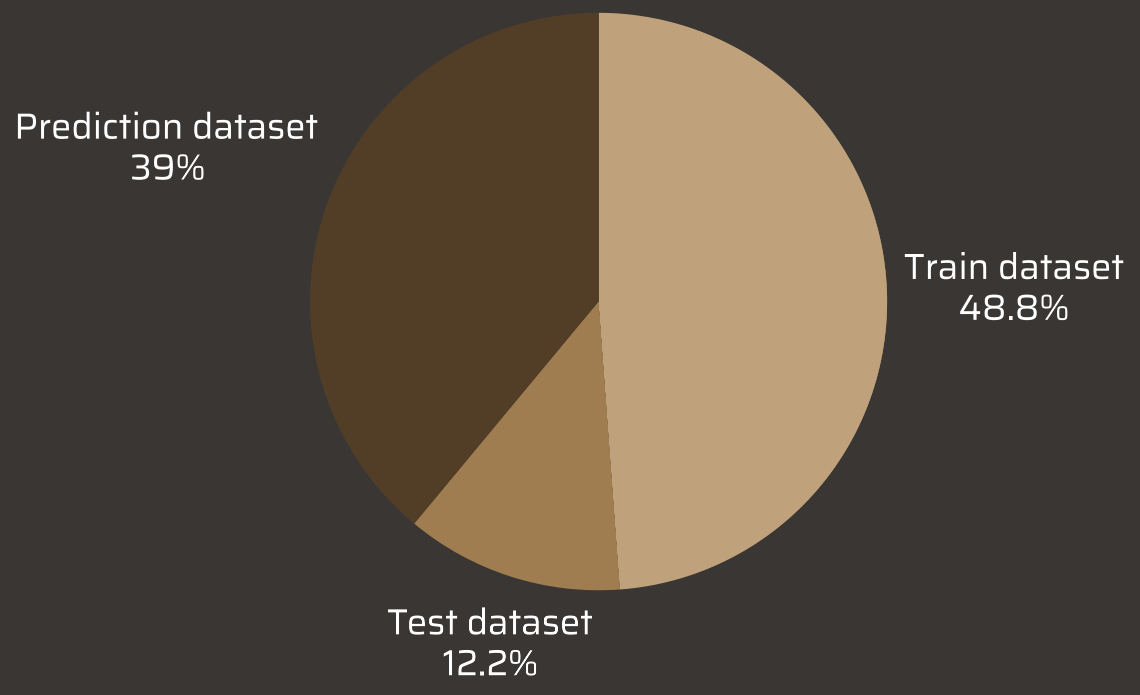
# MODEL BUILDING

## Build Models

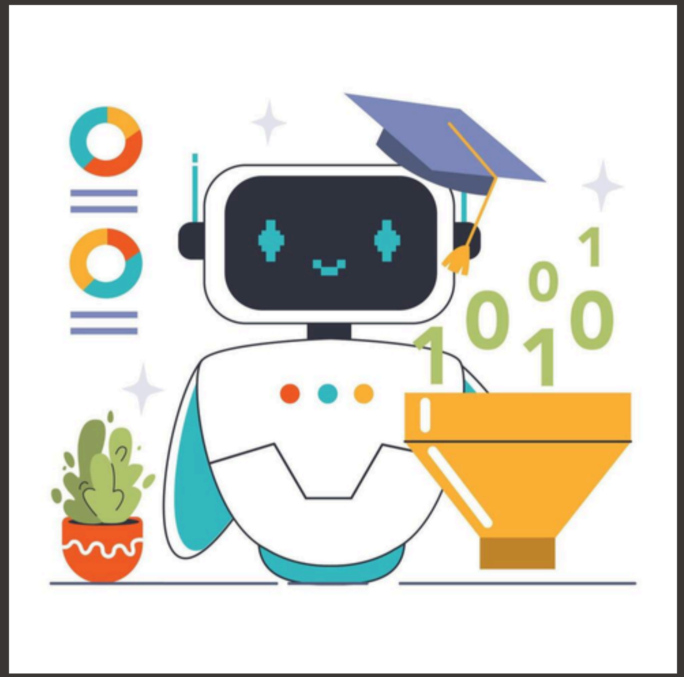
Choose features may affect customer demands




Encodes categorical labels into integer values.




Split data



Train models

# FINAL RESULTS

## Model Comparisions - RandomizedSearchCV

| Model | RMSE | MSE | TrainingTime | Memory Usage (MB) |
|---|---|---|---|---|
| Ridge | 0.36139 | 0.13060 | 1.9744 | 137.06 |
| ElasticNet | 0.36145 | 0.13065 | 1.9912 | 137.28 |
| SVR | 0.36162 | 0.13077 | 1.9796 | 136.17 |
| RandomForestRegressor | 0.36776 | 0.13524 | 1.6031 | 131.00 |
| GradientBoostingRegressor | 0.37403 | 0.13989 | 1.3402 | 132.65 |

# FINAL RESULTS

## Use Best Model To Predict Unrating Data

Best model

**Ridge Regression Model**

| Price | Trademark | Country | General_function | Predicted Rating |
|---|---|---|---|---|
| 390000.0 | 375 | 23 | 0 | 4.947837 |
| 22400.0 | 214 | 38 | 3 | 4.920278 |
| 151200.0 | 290 | 38 | 0 | 4.921739 |
| 39000.0 | 14 | 38 | 0 | 4.900257 |
| 39000.0 | 14 | 38 | 0 | 4.900257 |
| ... | ... | ... | ... | ... |
| 276000.0 | 208 | 38 | 14 | 4.861462 |
| 276000.0 | 342 | 38 | 4 | 4.885514 |
| 276000.0 | 342 | 38 | 15 | 4.934686 |
| 276000.0 | 234 | 7 | 5 | 4.946959 |
| 276000.0 | 157 | 8 | 15 | 4.873883 |

# THANK YOU FOR YOUR ATTENTION